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Neo-Cebu 2026 Security Framework: A Context-Aware Virtual Learning Environment with QR-Based Authentication for Public Education in Cebu City

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Chapter I

INTRODUCTION AND PROBLEM LANDSCAPE

Introduction

The vision for Neo-Cebu in 2026 demands a highly resilient, fully digitized public education infrastructure. As physical classrooms transition toward continuous virtual learning, reliance on web-based platforms for content delivery, grading, and student collaboration becomes absolute. The integrity of these digital spaces is essential to foster critical thinking and maintain rigorous academic standards. However, this rapid digitization introduces a complex attack surface. Traditional security measures that rely heavily on static passwords and implicit trust are no longer sufficient to protect the digital identities of students and educators within local academic networks (Fortinet, n.d.). In modern academic environments, students frequently suffer from "password fatigue," leading to the use of weak, easily compromised credentials that invite cyberattacks (Okta, 2025).

Target Users

Primary Users:

- Cebuano Students, High school and college students requiring a secure, frictionless, and password-less web platform to access digital learning.
- Educators and Teachers, Faculty members responsible for generating authentication codes, provisioning student access, and managing virtual classrooms.

Secondary Users:

- School Administrators, T staff monitoring access logs, managing educator accounts, and ensuring strict data privacy compliance for academic records.
- Local Government Units (LGUs) and DepEd Cebu, Authorities overseeing the broader public education infrastructure and smart-city security compliance.

Statement of the Problem

This study seeks to address the critical cybersecurity vulnerabilities and severe authentication friction facing the public education infrastructure of Neo-Cebu as it transitions to a fully digitized virtual learning environment by the year 2026. As digital academic systems become increasingly interconnected and automated, they are continually exposed to escalating cyber threats, including unauthorized access, data manipulation, and sophisticated AI-based attacks. Furthermore, reliance on traditional login credentials exacerbates these risks, as human-related factors—such as weak passwords, poor authentication practices, and general password fatigue—are identified as the weakest points in system security and account for a vast majority of breaches. Consequently, without a modernized defense framework, these unmitigated vulnerabilities expose the educational network to data interception, adversarial AI manipulation, and unauthorized control, directly threatening the fundamental integrity, confidentiality, and availability of student records and public education in Cebu City.

Specific Questions or Sub-Problems

1. How can QR-based authentication tied to physical student IDs eliminate password fatigue and mitigate AI-driven credential harvesting?

2. In what ways can a Context-Aware Zero Trust Architecture (utilizing JSON Web Tokens) prevent unauthorized access and Deepfake identity spoofing within live virtual classrooms?
3. How effective is an aggressive, automated 10-minute server-side session termination protocol in preventing session hijacking in shared public learning spaces?
4. What mechanisms are required to ensure persistent, secure peer-to-peer collaboration channels against coordinated digital disruptions?

Scope and Boundaries

To avoid overgeneralization, the scope of this study is specifically limited to the development of a secure, web-based virtual learning platform utilizing the ASP.NET Core framework for public education infrastructure within Cebu City. The research focuses exclusively on software-level access controls, including Context-Aware Zero Trust, teacher-provisioned QR credentials, and encrypted virtual classrooms. The project boundaries explicitly exclude the implementation of physical security hardware (such as biometric turnstiles or CCTV), integration with broader smart-city municipal networks, and connections to live, non-educational government databases.

CHAPTER II

ATTACK SURFACE MAPPING

Attack Surface Mapping refers to the systematic identification and analysis of all possible entry points where a system can be exploited. For the proposed Context-Aware Virtual Learning Environment in Neo-Cebu, the attack surface encompasses the physical endpoints, the ASP.NET Core software architecture, web-based communication networks, and the human interactions of students and educators. As educational platforms become heavily reliant on continuous cloud-based data exchange, they are increasingly exposed to cybersecurity threats. Evaluating these vulnerabilities is critical to ensuring the integrity of the zero-trust architecture.

Hardware and Physical Vulnerabilities

While the proposed system is primarily a web-based application, it relies heavily on physical components at the endpoint level, specifically the devices used by students and the physical school IDs containing the authentication QR codes.

1. Theft, loss, or unauthorized duplication of the physical student ID containing the QR code. Related Study, Al-Riyami et al. (2023) demonstrated that physical QR codes utilized for authentication are highly susceptible to "shoulder surfing" and unauthorized high-resolution duplication, requiring secondary contextual checks to prevent misuse.

2. Compromised or unsecured student endpoints (e.g., malware-infected personal laptops or mobile phones). Related Study, Research indicates that Bring Your Own Device (BYOD) policies in educational settings significantly expand the hardware attack surface, making malware-infected student endpoints a primary vector for network infiltration (EDUCAUSE, 2024).

3. Physical tampering with shared public terminals (e.g., internet cafes) used to access the learning environment. Related Study, Zhang et al. (2023) highlighted that public and shared computing hardware often harbor undetected keyloggers and screen-capturing software, allowing attackers to harvest credentials and monitor sessions without triggering network alarms.

Impact

- Unauthorized extraction of the QR authentication credential.
- Keylogging or screen-capturing of sensitive video lectures and chat logs.
- Circumvention of the system's software-level security due to hardware-level compromise.

Software Vulnerabilities

Software systems include the ASP.NET Core backend, the centralized database, the JSON Web Token (JWT) management system, and the integrated video and chat modules.

1. Insecure storage of JWTs in the browser exposing them to Cross-Site Scripting (XSS). Related Study, The OWASP Foundation (2023) identifies XSS as a critical threat to token-based authentication; if tokens are stored in LocalStorage rather than HttpOnly cookies, malicious scripts can easily extract them to hijack user sessions.

2. Unpatched vulnerabilities in third-party libraries used for the real-time video call implementation. Related Study: Sharma & Gupta (2022) found that modern web applications are highly vulnerable to supply-chain attacks, where unpatched third-party multimedia libraries are exploited to execute arbitrary code or bypass security protocols.

3. Application Logic Flaws that might allow a student to bypass the classroom-link isolation protocol. Related Study, Khan et al. (2024) emphasized that complex authorization logic, if improperly configured in the backend controller layer, can lead to Broken Access Control, allowing users to access restricted digital environments outside their assigned permissions.

Impact

- System takeover or unauthorized administrative control.
- Hijacking of active student sessions leading to data breaches involving academic records.
- Unauthorized injection of malicious payloads into the persistent classroom chatbox.

Network and Communication Vulnerabilities

The system depends on continuous, real-time communication between the client browsers, the ASP.NET Core servers, and the database to facilitate live video and persistent chat.

1. Utilization of unsecured, public Wi-Fi networks by students to access the platform. Related Study, The Cybersecurity and Infrastructure Security Agency (CISA, 2024) consistently warns that public access points lack proper encryption, allowing attackers on the same network to easily sniff traffic and capture unencrypted payloads.

2. Susceptibility to Man-in-the-Middle (MITM) attacks on the WebSocket connections powering the live chat. Related Study, Patel et al. (2023) emphasized that continuous data streams, such as those required for live virtual classrooms and persistent WebSockets, are prime targets for MITM interception and require rigorous Transport Layer Security (TLS).

3. Distributed Denial-of-Service (DDoS) attacks targeting the school's central server infrastructure. Related Study, Cloudflare's Global Threat Report (2024) noted a massive surge in DDoS attacks targeting the education sector, specifically aiming to overwhelm servers and disrupt synchronous digital learning environments.

Impact

- Interception of sensitive data, including JWTs and private peer-to-peer communications.
- Complete system downtime preventing students and teachers from accessing virtual classrooms.
- Degradation of video call quality and disruption of real-time collaboration.

Human Factor Vulnerabilities

Despite robust technical controls, human interaction remains one of the weakest points in the system's security architecture.

1. Students leaving their authenticated sessions open and unattended in shared spaces. Related Study, Recent analyses of "Hydra-Pulse" rapid-extraction threats show that automated bots can exploit an unattended, active terminal to harvest massive amounts of data in the short window before a standard session timeout triggers (arXiv, 2026).

2. Educators accidentally sharing a student's QR code or the isolated classroom link publicly. Related Study, The Verizon Data Breach Investigations Report (2023) states that human error and "misdelivery" (sending sensitive information to the wrong recipient or public forum) account for a significant percentage of all educational data breaches.

3. Susceptibility of teachers to targeted social engineering attacks aimed at granting unauthorized access. Related Study: Okta (2025) reported that educators are frequent targets of spear-phishing campaigns where attackers pose as IT administrators or students to trick teachers into provisioning unauthorized accounts or bypassing MFA protocols.

Impact

- Unauthorized, localized system access via unattended terminals.
- Data leakage of proprietary classroom materials.
- Mismanagement of student digital identities.

Summary

Component	Vulnerability Example	Impact
Hardware	Duplicated QR IDs, infected endpoints	Credential theft, unauthorized entry
Software	XSS targeting JWTs, API flaws	Session hijacking, data breaches
Network	MITM, DDoS on web sockets	Communication failure, data interception
Human	Unattended terminals, social engineering	Unauthorized localized access

CHAPTER III

PROPOSED INNOVATIVE SOLUTION

Rationale for Selecting the Solution

The attack surface mapping in Chapter II highlighted critical vulnerabilities across endpoints, network layers, and human interactions. These weaknesses expose the Neo-Cebu virtual learning environment to automated credential harvesting, session hijacking, and Deepfake identity spoofing. Conventional authentication methods relying on static, student-generated passwords are no longer viable, as human error and password fatigue frequently compromise system integrity (Okta, 2025).

To address these challenges, this study proposes a Context-Aware Zero Trust Architecture integrated with a QR-Based Authentication system, built upon the ASP.NET Core framework. This solution shifts the security paradigm from implicit trust to continuous verification. By eliminating passwords entirely and utilizing physical-to-digital QR credentials provisioned strictly by educators, the system directly mitigates the vulnerabilities identified in Chapter II:

Hardware & Human Risks, QR codes eliminate password fatigue and phishing susceptibility.

Software Exploitation, JWT-based Zero Trust prevents unauthorized lateral movement between isolated classroom links.

Network Interception, Aggressive 10-minute session timeouts and context-aware token validation mitigate the window for session hijacking.

Technical Framework

The proposed solution operates on three interconnected security pillars designed specifically for public education infrastructure:

A. **Centralized Identity Provisioning (RBAC),** Utilizing ASP.NET Core Identity, the system enforces strict Role-Based Access Control (RBAC). Students cannot autonomously create accounts. Instead, verified educators act as the sole identity providers, generating one-time authentication codes and QR credentials for their designated students. This centralized approach prevents anonymous sign-ups and external brute-force attacks (Microsoft, 2024).

B. **Physical-to-Digital Credentialing (QR Authentication),** The system integrates scannable QR codes tied to the students' physical school IDs. When scanned via the web application, the frontend transmits the encrypted payload to the ASP.NET Core backend. This creates a frictionless, password-less login experience that bridges physical identity with digital access, highly suited for the usability needs of younger students.

C. **Context-Aware Zero Trust Architecture (Continuous Verification),** Unlike traditional systems that trust a user once logged in, this framework assumes continuous compromise. Upon a successful QR scan, the backend issues a secure JSON Web Token (JWT). Every subsequent action—joining a video lecture or sending a chat—requires the JWT to be re-verified against contextual parameters:

   - **Endpoint Isolation:** The token is only valid for the specific virtual classroom link generated by the teacher.
   - **Aggressive Session Termination:** Automated, server-side session invalidation triggers after exactly 10 minutes of inactivity, terminating hijacked connections in shared public spaces (Palo Alto Networks, n.d.).

Comparative Analysis of Defense

The attack surface mapping revealed that public school networks are exposed to unique constraints: they require high security but must operate on limited hardware budgets with high user accessibility. To address these risks, three defense mechanisms were evaluated: Zero Trust + QR (Proposed), Post-Quantum Cryptography (PQC), and Biometric Authentication.

- **Zero Trust + QR (Chosen)**
  - Selected as the primary defense because it provides continuous verification while remaining highly accessible and cost-effective for students using standard webcams and physical IDs.
  - Mitigates phishing, password fatigue, and session hijacking without requiring expensive endpoint hardware.
- **Post-Quantum Cryptography (PQC)**
  - Provides extreme encryption resilience but introduces significant latency for real-time video streaming and is computationally heavy for standard student devices.
  - Rejected for primary authentication due to latency and hardware constraints, though TLS/SSL is maintained for transport.
- **Biometric Authentication**
  - Rejected due to extreme privacy concerns regarding the storage of minor students' biometric data and high implementation costs.

Risk Mitigation

While the proposed system is highly secure, specific operational risks must be managed:

A. **Risk:** Loss or theft of the physical QR school ID.
   - **Mitigation:** The system allows educators to instantly revoke the compromised JWT and regenerate a new QR payload via the administrative dashboard, immediately locking out the stolen credential.

B. **Risk:** Cross-Site Scripting (XSS) attacks aiming to steal the JWT from the browser.
   - **Mitigation:** The ASP.NET Core backend is configured to store JWTs strictly in HttpOnly, Secure cookies, rendering them inaccessible to client-side JavaScript, paired with strict Anti-Forgery Token validation (OWASP, 2023).

C. **Risk:** Teacher account compromise leading to unauthorized student provisioning.
   - **Mitigation:** All educator and administrator accounts are protected by mandatory Multi-Factor Authentication (MFA), adding a secondary verification layer before account generation is permitted.

CHAPTER IV

IMPLEMENTATION ROADMAP

The implementation roadmap outlines a structured and phased deployment strategy for the Neo-Cebu Context-Aware Virtual Learning Environment. The primary goal of this rollout is to introduce the new QR-based authentication and Zero Trust architecture incrementally without disrupting existing live educational services. By utilizing a parallel deployment strategy, public schools can maintain their current Learning Management Systems (LMS) while slowly transitioning faculty and students to the secure ASP.NET Core platform. A phased deployment approach is widely recommended for critical public infrastructure to minimize operational risks and maintain service continuity (NIST, 2023).

Development Plan

This phased rollout systematically addresses the vulnerabilities identified in Chapter II by ensuring secure credential distribution and rigorous token validation testing. It supports the transition to a Zero Trust architecture with zero disruption to live classes, ensuring long-term efficiency, security, and scalability for Cebu City's public education network.

Phase 1, Infrastructure & Credential Preparation

- Provision cloud servers and establish the ASP.NET Core hosting environment and centralized database.
- Develop the administrative dashboard for centralized identity provisioning.
- Print and distribute the first batch of physical school IDs embedded with the secure QR codes to a select pilot group of students.

Phase 2, Security & Authentication Integration

- Implement the Context-Aware Zero Trust Architecture, specifically configuring the JSON Web Token (JWT) generation and validation pipeline.
- Enforce strict Role-Based Access Control (RBAC) rules for educator accounts.
- Code and test the automated 10-minute server-side session termination protocol.
- Secure WebSocket connections for the persistent chatbox using Transport Layer Security (TLS).

Phase 3, Core Application Deployment (Staging)

- Deploy the encrypted real-time video call modules and classroom hubs to a staging environment.
- Conduct internal penetration testing targeting the QR login flow to ensure tokens cannot be hijacked via Cross-Site Scripting (XSS).

Phase 4, Parallel Pilot Testing

- Deploy the system to a localized pilot group (e.g., select IT classes at Cebu Eastern College) while keeping the legacy LMS active.
- Run the new system alongside the existing educational platforms, allowing teachers to use both simultaneously.
- Monitor system performance, video streaming latency, and the friction of the QR-based login process for students.

Phase 4, Continuous Monitoring and Full Integration

- Gradually onboard additional public schools into the centralized system.

- Monitor active JWT sessions and endpoint access logs in real-time.
- Perform regular security patching on ASP.NET Core dependencies and third-party video libraries.

Rollback and Failover Mechanism

To ensure zero downtime during the transition, the system includes:

- **Automatic Rollback,** If the QR authentication server experiences an outage, educators are instructed to temporarily fall back to the legacy LMS (e.g., Google Classroom) without losing instructional time.
- **Failover Servers**, Redundant ASP.NET Core server nodes deployed behind a load balancer ensure that if one server crashes, traffic is automatically routed to a backup node to maintain continuous operation.
- **Database Replication,** Synchronized backups of chat logs and course materials prevent data loss during deployment updates.

This ensures service continuity and reliability for both students and educators throughout the Neo-Cebu transition (Patel et al., 2023).

Phase	**Description**	**Duration**
Phase 1	Infrastructure & Credential Setup	1-2 months
Phase 2	Security & Authentication Integration	2 months
Phase 3	Core Application Deployment (Staging)	1 month
Phase 4	Parallel Pilot Testing	3 months
Phase 5	Full Integration & Optimization	Continuous

CHAPTER V

RISK ASSESSMENT & CIA TRIAD COMPLIANCE

This chapter evaluates how the proposed Context-Aware Virtual Learning Environment ensures Confidentiality, Integrity, and Availability (the CIA Triad)—the core principles of information assurance. Given the critical vulnerabilities and "Hydra-Pulse" threats identified in Chapter II, implementing strict CIA compliance within the ASP.NET Core architecture is essential to securing the Neo-Cebu public education infrastructure against advanced cyber threats (Sharma & Gupta, 2022).

Confidentiality

Confidentiality ensures that sensitive student data, academic records, and private communications are protected from unauthorized access or exposure.

Risks Identified

- Data interception via Man-in-the-Middle (MITM) attacks on public school Wi-Fi.
- Unauthorized access to virtual classrooms using hijacked sessions or Deepfake identity spoofing.
- Compromise of student accounts due to automated credential harvesting.

Solutions Implemented

- **Context-Aware Zero Trust Architecture,** Continuous JSON Web Token (JWT) verification ensures that only authenticated, active tokens tied to specific classroom links can access video feeds and chat logs.
- **QR-Based Authentication,** Eliminates reliance on passwords, ensuring that credentials cannot be harvested via traditional phishing or brute-force AI attacks (Okta, 2025).
- **Transport Layer Security (TLS),** Encrypts all data in transit, including live WebSocket chat streams and API calls to the centralized database.

Outcome

- Secure student-to-teacher communication.
- Robust protection of Personally Identifiable Information (PII) and academic records.
- Prevention of unauthorized entry into live virtual classrooms.

Integrity

Integrity ensures that academic data, chat payloads, and system configurations remain accurate, consistent, and unaltered by unauthorized parties.

Risks Identified

- Malicious modification of grades or course materials via SQL injection or application logic flaws.
- Cross-Site Scripting (XSS) attacks injecting malicious payloads into the persistent classroom chatbox.
- Unauthorized provisioning of student accounts by non-educators.

Solutions Implemented

- **Role-Based Access Control (RBAC),** Strict permissions enforced by ASP.NET Core Identity guarantee that only verified educators have the authority to generate or alter student QR credentials and classroom links.
- **Cryptographic Token Signatures,** JWTs are cryptographically signed by the backend server; any tampering with the token payload immediately invalidates the session (OWASP, 2023).
- **Entity Framework Core Defenses,** Utilizing parameterized queries through the .NET Object-Relational Mapper (ORM) to completely neutralize SQL injection vulnerabilities.

Outcome

- Highly accurate and tamper-proof academic data.
- Reliable and safe peer-to-peer collaboration environments.
- Strictly controlled identity management.

Availability

Availability ensures that the virtual learning environment and its critical services always remain accessible and operational for students and educators, especially during peak school hours.

Risks Identified

- Distributed Denial-of-Service (DDoS) attacks designed to overwhelm the school's central servers.
- System downtime during software updates or deployment phases.
- disruptions severing the connection between the client browser and the ASP.NET Core backend.

Solutions Implemented

- **Redundant Server Architecture,** Deployment of multiple ASP.NET Core nodes behind a load balancer to distribute traffic and prevent single points of failure.
- **Deployment Strategy,** As outlined in Chapter IV, maintaining legacy Learning Management Systems (LMS) as a fallback mechanism ensures zero downtime during the initial rollout.
- **Aggressive Session Management,** The 10-minute automated timeout not only prevents session hijacking but also frees up server resources by terminating inactive WebSocket connections, optimizing overall bandwidth.

Outcome

- Continuous, uninterrupted virtual classroom operations.
- Minimal latency during high-volume login periods (e.g., the start of the school day).
- Reliable educational services for Cebu City commuters and remote learners.

CIA Triad Summary Table

Security Principle	Threat Example	Solution	Result
Confidentiality	Credential theft, MITM	QR Auth, TLS, Zero Trust JWTs	Secure access & data privacy
Integrity	Data tampering, XSS	RBAC, Token Signatures, ORM	Accurate records & safe chat
Availability	DDoS, Server crashes	Load balancing, Session optimization	Continuous system operation

By integrating a password-less QR authentication system with a Context-Aware Zero Trust Architecture, the proposed ASP.NET Core solution systematically dismantled the high-risk attack vectors associated with traditional, static-password systems. This framework significantly reduces the threat of AI-driven credential harvesting, strengthens system resilience against session hijacking, and ensures secure, reliable digital learning. This compliance aligns with modern information assurance standards, providing a secure foundation for the Neo-Cebu 2026 educational paradigm (Khan et al., 2024).